

TerrAgro: Soil Classification Using Machine Learning

Case Study 139 | Machine Learning Fundamentals & Final Project Evaluation

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Final Project Title: TerrAgro: An Intelligent Soil & Agronomic Intelligence System Using Supervised Machine Learning

Course: Machine Learning Fundamentals (CS401 / ML202)

Assigned Problem Domain: Case Study 139: Soil Classification Using Machine Learning

Executive Summary & Quick Navigation

TerrAgro is an end-to-end Machine Learning precision agriculture platform that predicts optimal soil taxonomic classification and agronomic crop suitability from 9 fundamental geochemical and physical indicators: **pH, Nitrogen (N), Phosphorus (P), Potassium (K), Moisture (%), Organic Matter (%), Electrical Conductivity (EC), Temperature (°C), and Humidity (%)** across **8 distinct soil categories**.

Resource	File Link	Description
■ Academic Project Report	docs/markdown/PROJECT_REPORT.md / PDF	Full 16-chapter technical documentation formatted for 100-mark evaluation.
■ Viva Examination Guide	docs/markdown/VIVA_PREPARATION_GUIDE.md / PDF	25 high-yield viva questions, formulas, examiner traps, and oral strategy.
■ ML Topics Guide	docs/markdown/ML_TOPICS_EXPLAINED.md / PDF	Plain-English explanation of all ML concepts (HOW, WHY, WHEN).
■ Project Files Guide	docs/markdown/PROJECT_FILES_EXPLAINED.md / PDF	Comprehensive file-by-file breakdown.
■ ML Models Guide	docs/markdown/ML_MODELS_EXPLAINED.md / PDF	Deep dive into the 5 ML models (Logistic Regression, KNN, DT, RF, GB).
■ Jupyter Notebook	Soil Classification Case Study 139.ipynb	Complete, executed interactive Jupyter notebook with inline plots.
■ Streamlit Web Application	app.py	Interactive web dashboard running locally at <code>_CODE_PH_0_</code> .
■ Training Pipeline Script	train_and_evaluate.py	Reproducible script generating models, EDA plots, and benchmark CSVs.
■ Dataset Directory	dataset/	2,800 balanced soil observations and batch upload test templates.

Project Architecture & Workflow

[Problem Formulation & Case Study 139 Objectives]

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[Dataset Ingestion: 2,800 Observations across 8 Soil Orders]

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[Exploratory Data Analysis: Correlation Heatmap, Boxplots, NPK Distributions]

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[Preprocessing Impact Benchmark: Raw vs. StandardScaler Normalized]

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[Model Training & 5-Fold Stratified CV: LR, KNN, DT, RF, GB]

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[Comparative Evaluation: Accuracy, Macro F1, Recall, Confusion Matrix]

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[Model Serialization (joblib) & Feature Importance Analysis]

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[Streamlit Deployment: Live Classifier, Batch CSV, Advisory Hub, Viva Hub]

■ Quantitative Model Benchmark Leaderboard

Machine Learning Algorithm	Test Accuracy	Macro Precision	Macro Recall	Macro F1-Score	Weighted F1-Score	5-Fold Stratified CV	Preprocessing Applied
■ Logistic Regression	99.29%	99.29%	99.29%	99.29%	99.29%	98.54 ± 0.86%	StandardScaler
■ Random Forest Classifier	99.11%	99.11%	99.11%	99.11%	99.11%	98.50 ± 0.58%	None (Scale Invariant)
■ Gradient Boosting Classifier	98.39%	98.44%	98.39%	98.40%	98.40%	98.36 ± 0.51%	None (Scale Invariant)
4■■ K-Nearest Neighbors (KNN)	98.04%	98.06%	98.04%	98.03%	98.03%	98.11 ± 0.54%	StandardScaler (+5.54% gain)
5■■ Decision Tree Classifier	95.89%	95.91%	95.89%	95.88%	95.88%	95.68 ± 0.63%	None (Scale Invariant)

■ Key Research Questions Answered (Case Study 139)

1. Can soil type be automatically classified?
Yes (> 99% accuracy). Machine learning models readily capture geochemical and physical soil fingerprints across 8 orders.
2. Which soil parameter contributes most to classification?
Electrical Conductivity (18.84%) and Organic Matter (17.56%), followed by Moisture (16.10%) and pH (10.14%).
3. Which algorithm provides the highest F1-score?
Logistic Regression (99.29%) and Random Forest (99.11%).
4. Which soil categories are frequently confused?

Sandy Loam vs. Red Soil (minor overlap due to shared low moisture and low organic matter).

5. Does preprocessing improve classification?

Yes! KNN gained **+5.54% accuracy** (from 92.50% to 98.04%), and Logistic Regression achieved smooth gradient convergence.

6. Can the model classify a completely new soil sample?

Yes. Serialized pipeline processes arbitrary raw test inputs in real time with confidence estimates.

■ Installation & Local Execution

1. Clone & Navigate to the Project Directory

```
cd "/Users/rayhaansiledar/Desktop/ML Final Project"
```

2. Install Dependencies

```
python3 -m pip install -r requirements.txt
```

3. Run the Training Pipeline (Optional - Pretrained models already included)

```
python3 train_and_evaluate.py
```

4. Launch the Interactive Streamlit Web Application

```
streamlit run app.py --server.port 8503
```

Open your browser at `_CODE_PH_0_`.

■ Organized Project Structure

```

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■■■■ app.py # Main Streamlit Web Application
■■■■ train_and_evaluate.py # Master Training & Benchmark Pipeline
■■■■ Soil_Classification_Case_Study_139.ipynb # Interactive Jupyter Notebook
■■■■ requirements.txt # Dependencies file
■■■■ README.md # Main project overview documentation
■
■■■■ docs/ # ■ All Documentation & PDFs
■ ■■■■ markdown/ # Markdown source files
■ ■ ■■■■ PROJECT_REPORT.md # Formal 16-Chapter Academic Report
■ ■ ■■■■ VIVA_PREPARATION_GUIDE.md # 25+ Viva Voce Q&As
■ ■ ■■■■ ML_TOPICS_EXPLAINED.md # ML Concepts (HOW, WHY, WHEN)
■ ■ ■■■■ PROJECT_FILES_EXPLAINED.md # File-by-File Breakdown
■ ■ ■■■■ ML_MODELS_EXPLAINED.md # 5 ML Models Deep Dive
■ ■■■■ pdf/ # Exported Publication-Grade PDFs
■ ■■■■ PROJECT_REPORT.pdf
■ ■■■■ VIVA_PREPARATION_GUIDE.pdf
■ ■■■■ ML_TOPICS_EXPLAINED.pdf
■ ■■■■ PROJECT_FILES_EXPLAINED.pdf
■ ■■■■ ML_MODELS_EXPLAINED.pdf
■ ■■■■ README.pdf
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■■■■ dataset/ # ■ Raw & Processed Soil Data
■ ■■■■ soil_classification.csv # 2,800 Multi-Parameter Observations
■ ■■■■ online_raw_soil_measures.csv # Live downloaded online agricultural dataset
■ ■■■■ test_samples.csv # 25-Sample Test Batch CSV Template
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■■■■ models/ # ■ Serialized Models & Benchmarks
■ ■■■■ logistic_regression.pkl # Trained Logistic Regression
■ ■■■■ knn.pkl # Trained KNN Model
■ ■■■■ decision_tree.pkl # Trained Decision Tree Model
■ ■■■■ random_forest.pkl # Trained Random Forest Ensemble
■ ■■■■ gradient_boosting.pkl # Trained Gradient Boosting Model
■ ■■■■ scaler.pkl # Trained StandardScaler
■ ■■■■ label_encoder.pkl # LabelEncoder mapping classes
■ ■■■■ model_metadata.json # Full project metadata & metrics
■ ■■■■ benchmark_comparison.csv # Comparative leaderboard CSV
■ ■■■■ preprocessing_impact.csv # Preprocessing impact data
■ ■■■■ feature_importance.csv # Gini & Boosting feature weights
■ ■■■■ confused_soil_pairs.csv # Error analysis class pairs
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■■■■ eda_plots/ # ■ 7 High-resolution EDA charts
■■■■ model_metrics/ # ■ 6 Evaluation & benchmark plots
■■■■ scripts/ # ■■ Helper & Utility Scripts
■■■■ convert_docs_to_pdf.py # Markdown-to-PDF batch converter

```

■ Authors & Academic Credits

- **Project Title:** Soil Classification Using Machine Learning (Case Study 139)
- **Academic Program:** B.Tech Computer Science & Engineering / Data Science & AI
- **Evaluation Criteria:** 100-Mark Project & Viva Voce Examination